

# Homework 1 - The Learning Procedure

## Instructions

Due 25 September at 11pm

### Instructions

- Use the space inside of

```
::: {.solbox data-latex=""}
```

```
:::
```

to answer the following questions.

- Do not move this file or copy it and work elsewhere. Work in the same directory.
- Use a new branch named whatever you want. Create it now! Can't come up with something, try here. Make a small change, say by adding your name. Commit and push now!
- Try to Knit your file now. Are there errors? Fix them now, not at 11pm on the due date.
- There MUST be some text between `::: {.solbox}` and the next `:::` or this will fail to Knit.
- If your code or figures run off the edge of the `.pdf`, you'll lose 2 points automatically.
- Be sure to add your name in the `author` field at the top.

### Generative AI Self Report

If you use any generative AI tool on homeworks or labs, indicate here how you used the tools and how they contributed to your learning/ability to understand the assignment. Include every prompt that you used.

This self report will not affect your grade in any way. Please be as honest as possible.

- your response here \*

# 1 Forecasting COVID in BC

In the course `Stat406` package, you will find COVID-19 case data for the five health authorities in BC. We're going to play with a few models for making forecasts. Your goal is to **predict 2-week ahead case counts for all 5 health authorities**.

There are 2 BC Covid datasets we need for this assignment: `bccovid_train` and `bccovid_test`. Both have 6 variables: `HA`, `date`, `cases+14`, `cases+0`, `cases-7`, `cases-14`. `cases+14` is the response: it is the total number of cases observed in that specific HA, in the week ending 14 days after `date`. The other `cases` variables are features: the total number of cases in the week ending on the date, those for 1 week earlier and those for 2 weeks earlier. For complete descriptions, try `?bccovid` after loading the course package (or examine the package documentation on the web).

## 1.1 (1 point)

Using only the training data: a. (0.5 point) make a plot of today's cases over time. colour the points by health authority. b. (0.5 point) Write a few sentences discussing the data. Describe any similarities or differences across health authorities you notice. How do cases seem to “evolve” over time?

**Hint:** to use these covariates in `ggplot()`, you have to put them inside “backtick” marks, like ``cases+0``.

- a.
- b.

## 1.2 (2 points)

Estimate a linear model of 2-week ahead cases (`cases+14`) on the three other case features using the training data.

- (0.25 points) Plot the residuals against time (again coloured by HA).
- (0.5 points) Make a QQ-plot of the residuals (coloured by HA). Do these plots seem support the standard assumptions for a linear model? Why or why not?
- (0.5 point) Report the slope coefficient on `cases+0`. Describe what this coefficient means in the context of this problem. Does this interpretation make sense, why or why not?
- (0.75 points) Does this model seem appropriate? Justify your answer.

**Hint:** to use these covariates in a formula, you have to put them inside “backtick” marks, like ``cases+0``.

- a.
- b.
- c.
- d.
- e.

### 1.3 (2 points)

Examine the `bcpop` data set. Using an appropriate `*_join()`, add this column to the training data. Now scale all the columns by the population to get “cases per capita”. Multiply by 100,000 just to get reasonable looking numbers.

- (0.5 point) Make a plot of today’s cases over time scaled by population (and coloured by HA). Furthermore, scale the y-axis on the `log10()` scale. (Hint: try `?scale_y_log10`). How does this compare to the plot in Question 1.1? Write 2 complete sentences.
- (1.5 points) Repeat Question 1.2 with both your scaled data after taking `log()` of all the predictors AND using HA as a categorical predictor.

**First bullet.**

**Repeat Q2 with different data.**

- a.
- b.
- c.
- d.
- e.

### 1.4 (1 point)

Plot the fitted values from the two models against each other (fits from the Question 1.2 on the x-axis, with fits from Question 1.3 on the y-axis). Be sure to transform them as needed to be comparable. Colour the points by HA. What do you notice?

(placeholder text)

## 1.5 (1 point)

Calculate the root mean squared error of both models. Discuss the results in 2 sentences.

(placeholder text)

## 1.6 (1 point)

Use the K-fold CV function from class (loaded below) to compare these two models (the first model being the one with `bccovid_train` from Question 1.2, and the second being the one with the log of case rates and the factor for HA from Question 1.3). Use  $K = 10$ . In order to do this, you'll need to be sure that the errors are on the same scale.

Use `Cases` as the scale. This means you'll have to specify the model and the error function carefully.

- (0.5 points) What are the CV scores for both models?
- (0.25 points) Which do you prefer?
- (0.25 points) How much more accurate is it?

```
## @param data The full data set
## @param estimator Function. Has 1 argument (some data) and fits a model.
## @param predictor Function. Has 2 args (the fitted model, the_newdata) and
##   produces predictions
## @param error_fun Function. Has one arg: the test data, with fits added.
## @param kfolds Integer. The number of folds.
kfold_cv <- function(data, estimator, predictor, error_fun, kfolds = 5) {
  n <- nrow(data)
  fold.labels <- sample(rep(1:kfolds, length.out = n))
  errors <- double(kfolds)
  for (fold in 1:kfolds) {
    test.rows <- fold.labels == fold
    train <- data[!test.rows, ]
    test <- data[test.rows, ]
    current_model <- estimator(train)
    test$.preds <- predictor(current_model, test)
    errors[fold] <- error_fun(test)
  }
  mean(errors)
}
```

- a.
- b.
- c.

### 1.7 (1 point)

Use your preferred model to predict the test set. a. (0.25 points) Plot the actual values and the predictions, coloured by HA. Use lines for the predictions and dots for the observations. b. (0.25 points) For each HA, how accurate is the prediction on average (what is the mean absolute error on the scale of cases)? c. (0.25 points) What about overall? d. (0.25 points) How does this compare to the CV score you computed above?

- a.
- b.
- c.
- d.

## 1.8 (2 points)

- a. (0.25 points) Write the statistical model used for Question 1.3 (the log LM), and write the distributional assumption on the model residuals.
- b. (0.5 points) Assume we now modify the distribution on the residuals to be i.i.d. draws from Laplace distribution with  $\mu$  (location parameter) = 0 and  $b$  (scale parameter) = 1. I claim that this model is more robust to outliers in the data. Why? Limit your answer to two sentences. (Hint: Plot the Laplace distribution and the Normal distribution side-by-side, and compare the tail behavior and probability density for large deviations from the mean.)
- c. (0.5 points) Derive the MLE for this model. This estimator does not have a closed form, so your final answer should be expressed as an optimization problem (i.e.  $\operatorname{argmax}$  or  $\operatorname{argmin}$ ) over a summation, just like what we derived in class for logistic regression. Show your work for taking the derivative and setting it to zero, even though you cannot solve it explicitly.
- d. (0.25 points) What is the continuous risk function that, if we were to take an ERM perspective, yields the same estimator as the MLE?
- e. (0.5 points) Without actually fitting the model, give one (~2 sentence reason) why the Laplace model might perform better than the existing log LM for the COVID data. Give one (~2 sentence reason) why the existing model might perform better. Base your answers on the graphs and metrics you have computed so far.

- a.
- b.
- c.
- d.
- e.